

Thursday Warm-up: Potentials

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October 2, 2025

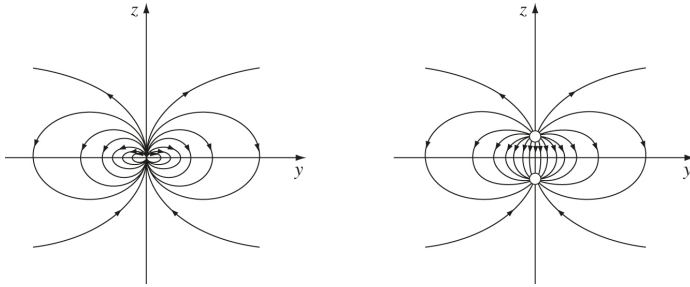


Figure 1: (Left) A dipole in the limit that $d \rightarrow 0$ and $q \rightarrow \infty$. (Right) A physical dipole.

Show that the dipole moment is independent of a shift in the origin, if the total charge is zero.

1 Memory Bank

- **Gradient** in spherical coordinates:

$$\nabla V = \frac{\partial V}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{\boldsymbol{\phi}} \quad (1)$$

- **Laplace's equation** in spherical coordinates, with azimuthal symmetry:

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) = 0 \quad (2)$$

- **Multipole expansion** in spherical coordinates:

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n P_n(\cos \alpha) \rho(\mathbf{r}') d\tau' \quad (3)$$

2 Multipole Expansion

1. Generate the approximate voltage of a physical dipole aligned with the z-axis ($n = 1$ in Eq. 3, and $P_1(x) = x$).
2. In analogy to the *moment* of a probability distribution function (PDF) let the **dipole moment** of a charge distribution be

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d\tau' \quad (4)$$
3. Formulate the expression for the voltage of the dipole in terms of the dipole moment. Note that $\hat{\mathbf{r}} \cdot \mathbf{r}' = \cos \theta$ if $\mathbf{p}$ is along the z-axis.
4. Now compute the **E**-field of a dipole using the gradient in spherical coordinates.
5. List the magnitude and direction of the **E**-field at the following points:
 - A point far from the origin, on the xy-plane.
 - A point far from the origin, on the z-axis.
 - A point far from the origin, on the -z-axis.
6. **Think conceptually.** What is the sum of the monopole and dipole terms in the multipole expansion for a charge distribution comprised of two anti-parallel dipoles?